

Material and Colored Illumination Separation from Single Real Image via Self-Supervised Domain Adaptation

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Abstract

Separating material and illumination from images allows for realistic manipulation, which is crucial for image editing and augmented reality. In this work, we propose a training strategy for intrinsic decomposition networks that combines unsupervised domain adaptation with self-supervised constraints and sparse-guided fine-tuning in the target domain. Through two complex optimization steps, even a lightweight CNN can achieve outstanding material and illumination separation in real single images. The final separation results and various realistic image manipulation applications demonstrate the effectiveness and potential of our proposed method.

CCS Concepts

• **Computing methodologies** → *Image manipulation*.

Keywords

intrinsic decomposition, illumination separation, augmented reality

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1 Introduction

Intrinsic image decomposition, which separates an image into material component (reflectance) and illumination component (shading), is a promising technique for realistic image editing and achieving consistent lighting in augmented reality [Meka et al. 2021]. However, it is a challenging inverse problem in computer graphics due to the complex entanglement of material and illumination.

Most methods simplify this task by assuming white illumination, which inaccurately assigns all color information to reflectance and strips illumination of its natural color. Meka et al. [2021] proposed a method accommodating colored illumination, but it assumes indirect reflections, limiting its effectiveness for direct



Figure 1: Comparison of material and illumination separation results and applications based on the proposed method.

colored light sources. Additionally, most methods are trained on synthetic datasets and fine-tuned on sparsely annotated real data, leading to poor generalization on real images [Li and Snavely 2018].

To overcome these limitations, we extend the traditional white illumination separation to a universally applicable colored illumination for entire scenes. We mainly introduce a broadly applicable semi-supervised domain adaptation training method, enabling lightweight CNN to achieve excellent material and illumination separation on real images. Using the separated material and illumination components, we achieved high-quality virtual appearance manipulation under various conditions.

2 Approach

2.1 Intrinsic Decomposition Network

Reflectance (R) and shading (S) are three-channel RGB images that retain the scene's semantic content despite their physical differences. Using this property, we design a lightweight U-shaped network with one encoder and two decoders. The encoder extracts features from the original image, while the decoders separately output R and S , enabling efficient intrinsic decomposition from a single image.

2.2 Semi-Supervised Domain Adaptation Training Strategy

Most intrinsic decomposition networks rely on synthetic datasets due to the scarcity of dense reflectance and shading labels in real images. However, the domain gap between synthetic and real images often results in poor performance on real data. To overcome this, we introduce unsupervised domain adaptation, allowing training on both synthetic and real images without dense labels. Unlike tasks like image segmentation or restoration, synthetic datasets for intrinsic decomposition are imperfect. For example, zero-value masks in reflectance and shading in certain light source positions

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can lead to reconstruction inconsistencies. Directly applying unsupervised domain adaptation could transfer these flaws to the target domain. To mitigate this, we propose a semi-supervised domain adaptation framework based on intrinsic assumptions, using a two-step training process.

Step 1. We treat the intrinsic decomposition network as the generator G and introduce two U-Net networks as discriminators D_S and D_R for S and R , respectively. The initial optimization goal is to minimize pixel-level supervised loss to train the generator on the synthetic source domain:

$$L_S^p = \frac{1}{N} \sum_{i=1} (\alpha_i - \alpha'_i)^2 + 3 * \frac{1}{N} \sum_{i=1} (\nabla \alpha_i - \nabla \alpha'_i)^2 \quad (1)$$

where i denotes the pixel index, N the total number of pixels, α_i the generated intrinsic images R and S , and α'_i their corresponding ground truth. Next, we train the discriminators D to distinguish between outputs from source and target images while simultaneously training G to deceive them. Specifically, L_{BCE} represents the Binary Cross Entropy Loss. The optimization objectives for the reflectance and shading discriminators are to minimize the following loss functions:

$$\begin{aligned} L_{D_R} &= \frac{1}{N} \sum_i L_{BCE}(D_R(R_s), 1) + \frac{1}{N} \sum_i L_{BCE}(D_R(R_t), 0) \\ L_{D_S} &= \frac{1}{N} \sum_i L_{BCE}(D_S(S_s), 1) + \frac{1}{N} \sum_i L_{BCE}(D_S(S_t), 0) \end{aligned} \quad (2)$$

The adversarial objective for optimizing the generator is to minimize the following loss function:

$$L_t^{DR} + L_t^{DS} = \frac{1}{N} \sum_i L_{BCE}(D_R(R_t), 1) + \frac{1}{N} \sum_i L_{BCE}(D_S(S_t), 1) \quad (3)$$

In the equation, R_s and S_s are the intrinsic images generated in the source domain, while R_t and S_t are those generated in the target domain. To prevent the generator from transferring undesirable styles from the source to the target domain, we introduce a semi-supervised reconstruction loss in the target domain during training to ensure the stability of R and S :

$$L_t^{recon} = \frac{1}{N} \sum_i (I_t - R_t * S_t)^2 \quad (4)$$

Finally, the generator's optimization objective in the first training step is: $\min_{\theta_G} L_S^p + L_t^{DR} + L_t^{DS} + L_t^{recon}$. For the discriminators, the optimization objectives are: $\min_{\theta_{D_R}} L_{D_R}$; $\min_{\theta_{D_S}} L_{D_S}$.

Training alternates between optimizing the generator and discriminator, using Hypersim [Roberts et al. 2021] as the source domain dataset and real images from SAW [Kovacs et al. 2017] as the target domain, over 50 epochs.

Step 2. Although real images lack dense intrinsic ground truth, sparse annotation masks can be manually created. These annotated regions align with intrinsic image priors and can be used for training. After the first step, we fine-tune the generator network using the annotated masks from the target domain SAW dataset. The loss

function for SAW is as follows:

$$\begin{aligned} L_{SAW} &= \sum_{j=R,G,B} \frac{1}{N_C} \sum_{i=1}^{N_C} \left(\log(S_i^j) - \frac{1}{N_C} \sum_i \log(S_i^j) \right)^2 \\ &+ 0.1 * \sum_{j=R,G,B} \frac{1}{N_E} \sum_{i=1}^{N_E} \left(\log\left(\frac{S_i^j}{I_i^j}\right) - \frac{1}{N_E} \sum_i \log\left(\frac{S_i^j}{I_i^j}\right) \right)^2 \end{aligned} \quad (5)$$

where j refers to the RGB channels, and i is the pixel index. N_C denotes manually annotated constant shading regions for enforcing consistency in S , while N_E represents shadow edges for consistency in inverse reflectance.

In step 2, we add these constraints from step 1 but remove the discriminator loss for S to avoid excessive shading smoothing. The second step focuses on optimizing the following three objectives: The first optimization objective is: $\min_{\theta_G} L_S^p + L_t^{DR} + L_t^{recon}$. The second optimization objective is: $\min_{\theta_{D_R}} L_{D_R}$. The third optimization objective is: $\min_{\theta_G} L_{SAW}$. During training, we alternate between these three objectives, with a total of 40 training epochs.

3 Results and Applications

In Figure 1, we compare the real illumination and material images generated by our method with those from existing methods, showing our approach produces higher-quality intrinsic images. We also demonstrate realistic virtual manipulations with impressive visual effects. Additional examples are provided in the supplementary materials.

4 Conclusion

We present a training strategy for intrinsic decomposition networks that closes the domain gap between synthetic and real images, enabling lightweight CNNs to achieve effective material and illumination separation on real images. Our method supports high-quality tasks like relighting and augmented reality, proving its utility for realistic virtual manipulation.

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